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AI-Driven Employee Retention Platform

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INTRODUCTION

The AI-Driven Employee Retention Platform is a web-based decision support system designed for Small and Medium Enterprises (SMEs). By integrating an explainable rule engine with machine learning, the system analyzes key performance metrics like task load and deadline adherence to deliver real-time "Need Help" alerts and periodic turnover summaries. This platform replaces traditional, manual supervision with automated, proactive intervention to preserve corporate talent and secure company stability.

PROBLEM STATEMENT

Employee retention management in startups and project-based SMEs relies on manual communication tools (e.g., WhatsApp) and subjective year-end evaluations. This decentralized approach causes delayed responses, biased performance evaluations, and unidentifiable employee burnout patterns until it is too late. The lack of automated predictive tracking leaves businesses vulnerable to high turnover and expensive replacement costs that compromise both organization culture and operational prosperity.

OBJECTIVES

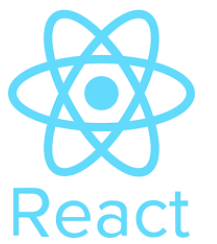
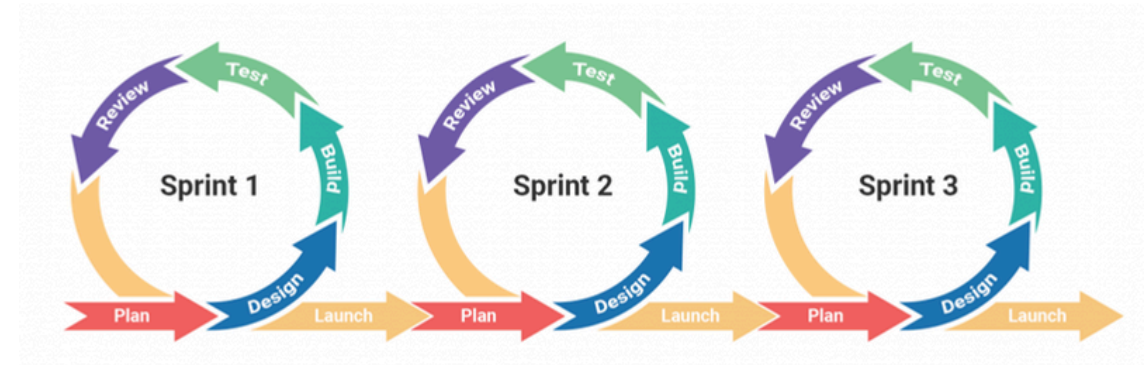
- To analyze existing SME management systems and identify key performance indicators correlated with employee Need Help and Turnover.
- To design and develop an AI-driven platform integrating a rule-based engine for alert detection and a Logistic Regression model for turnover prediction.
- To evaluate the predictive accuracy of the AI models and validate the platform's overall usability through User Acceptance Testing (UAT).

SCOPE

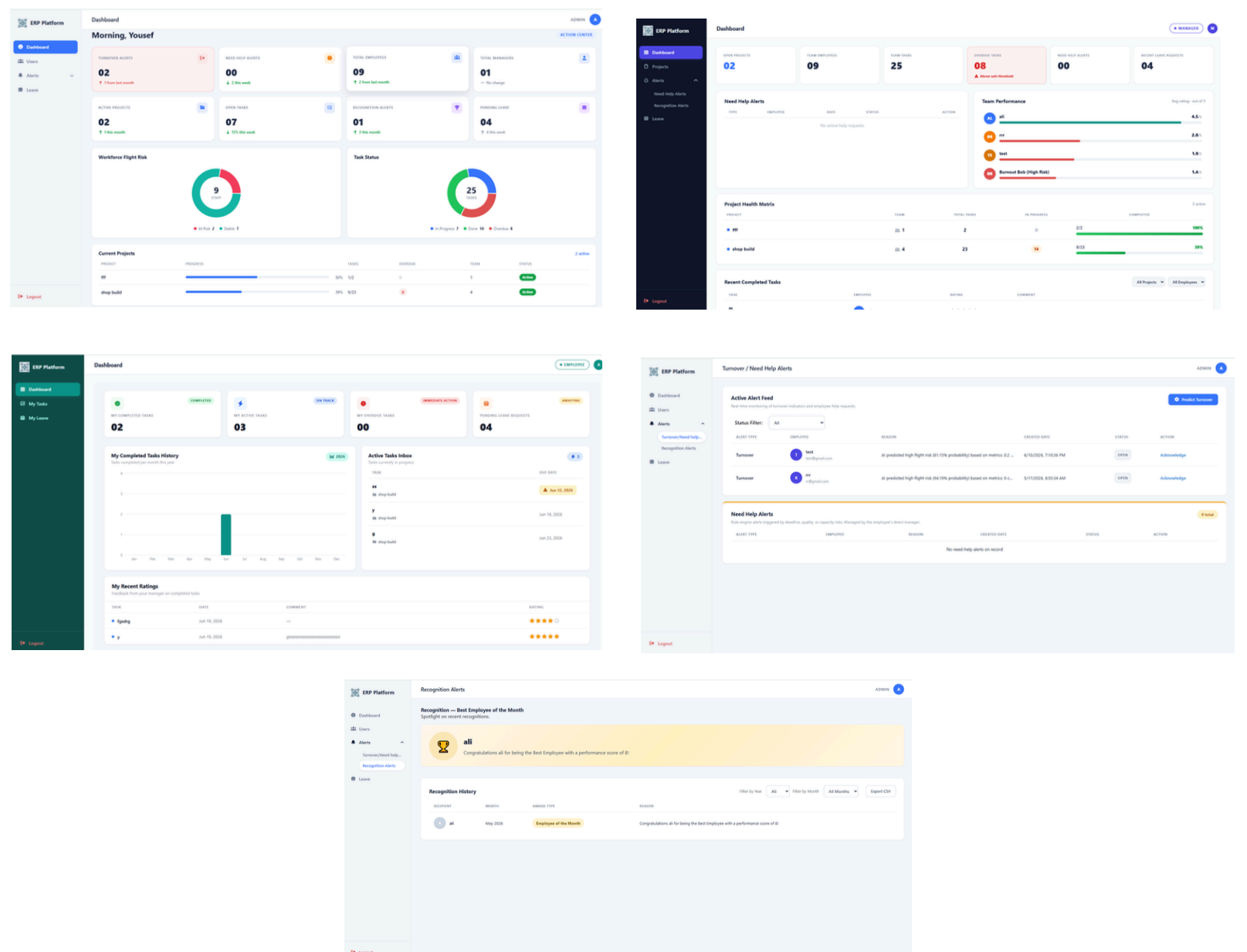
The platform is a web application tailored for a startup real estate tender company with approximately 5 employees, focusing exclusively on Dashboards, User management, Task, Performance, and Leave Management modules. The architecture consists of a React.js frontend, a Node.js (Express) backend, a MySQL database, and an isolated Python microservice executing a Logistic Regression algorithm trained on synthetic datasets. The system strictly serves three user roles: Admin/HR, Manager, and Employee.

METHODOLOGY

This project utilizes the Agile Software Development Life Cycle (SDLC) framework to ensure continuous improvement through iterative planning, development, and testing phases.



RESULT



CONCLUSION

The AI-Driven Employee Retention Platform successfully transforms SME human resource management from a delayed, subjective process into a proactive, data-driven system. By continuously monitoring real-time task workloads and operational metrics, the integrated AI models accurately detect early signs of employee burnout and predict flight risks before resignations occur. Ultimately, this platform eliminates bias, cuts turnover costs, and creates a healthier workplace.